MASTER'S THESIS DEFENSE / UNIVERSITY OF TRIESTE

Unsupervised State Bucketing

for Mixture-of-Experts in Resource-Constrained Chess Engines

Can a tiny chess engine choose the right expert?

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Specialization must preserve search speed

The deployment constraint

120 MHz	192 KB	512 KB
CPU frequency	Internal RAM	Internal flash

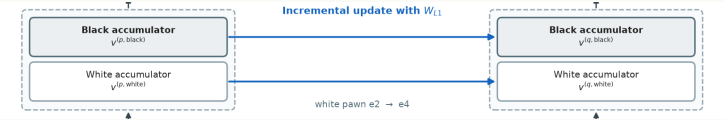
Alpha–beta search repeatedly calls the evaluation function. A slower evaluator can reduce search depth.

Can specialization improve prediction within this budget?

Target: Wio Terminal, ARM Cortex-M4. Hardware specifications, not deployment measurements. Thesis §4.8.

NNUE updates only the changed features

Efficiently updatable neural network





Before *e2–e4*



After *e2–e4*

A move changes only a few active input features.

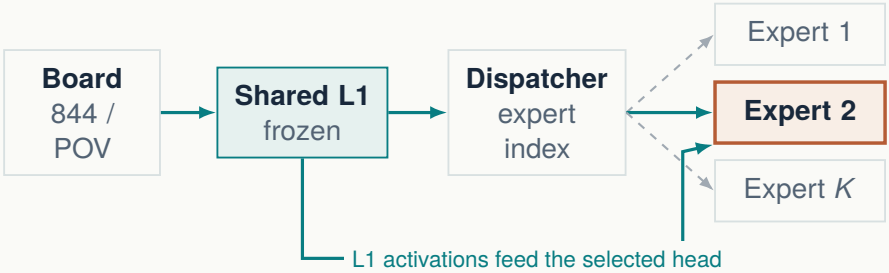
Subtract removed features.
Add new features.

The L1 accumulator persists across moves.

Crops of thesis Fig. 1.1. L1 denotes the shared first-layer representation throughout the presentation.

One accumulator, one selected expert

Hard mixture of experts



Fine-tuning duplicates the small heads. Only one head runs online.

Hard-routing architecture, thesis §§3.3–3.4. Multiple stored experts still consume additional memory.

Bucketing defines each expert's task

Alternative specialization signals

Bucket signal	Groups positions by	Available online?
Piece count	A hand-designed game-phase proxy	Yes, a fixed rule
L1 activations	Similar hidden representations	**Yes, already computed**
Sample gradients	Similar head-parameter updates	**No, requires teacher WDL**
Jointly learned gate	An end-to-end routing objective	Yes, after training

L1 describes the position. Gradients describe how the head should change.

Thesis §3.3. Per-sample head gradient: $g_i = \nabla_{\theta_{\text{head}}} \mathcal{L}(f(x_i), y_i)$, with teacher target y_i.

Training and evaluation protocol

Teacher distillation and held-out comparison

150M

Training positions, mainly Lichess

Lc0 supplies win / draw / loss probabilities. A compact NNUE learns to match them.

Reference NNUE: 174,723 parameters.

1.5M

Independent test positions

50,000 positions

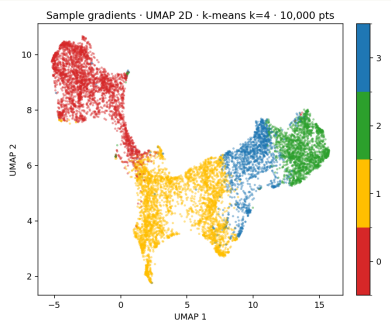
Held-out subset for the primary MoE comparisons.

Lower cross-entropy (CE) means closer to the teacher's WDL probabilities.

Thesis §§4.1–4.2, 5.1 and 5.5. CE does not directly measure chess Elo.

Gradients form more separated clusters

Offline structure



Silhouette score at $K = 8$

0.191 Gradients

0.003 L1

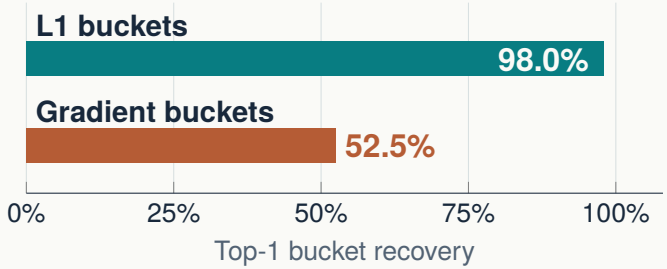
Gradient buckets capture a different signal from piece count and L1 geometry.

Separation alone does not establish better experts or reliable routing.

Thesis Fig. 5.3: gradient UMAP with $K = 4$. Table 5.2: silhouette at $K = 8$, measured in the respective representations.

Gradient buckets are hard to recover

Same L1 input, same MLP dispatcher, eight buckets



45.5
percentage-point gap

A target-dependent training signal is difficult to infer from L1 alone.

Thesis Tables 5.6–5.7: MLP $h = 64$, $K = 8$, 1M-position cache, 90/10 split. Transductive validation: clustering precedes the split.

Strong-base MoE gains are marginal

Primary experiments, test cross-entropy

Model	K	CE
Reference NNUE	–	**0.6303**
Piece-count MoE	8	0.6303
L1-clustered hard MoE	16	0.6300
Gradient-clustered hard MoE	8	0.6300
Soft-gated MoE, top-2	2	0.6304

-0.0003

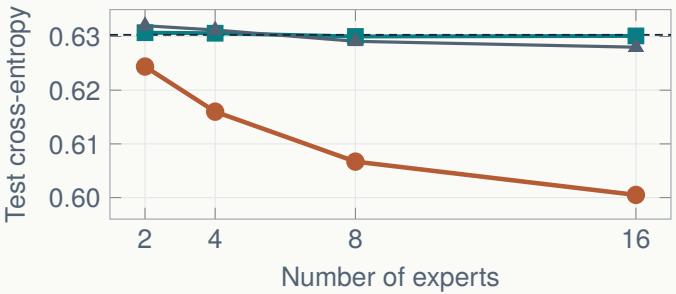
Best observed CE change

No decisive improvement over the reference NNUE.

Thesis Tables 5.10–5.11, 50,000 held-out positions. Lower CE is better. These results do not establish an Elo gain.

The oracle exposes assignment headroom

Same gradient-trained experts, different selection rules



At $K = 16$

Actual dispatcher
0.6301 CE

Gradient-centroid proxy
0.6280 CE

Minimum-loss oracle
0.6005 CE

Even the gradient-centroid rule falls far short of choosing the best expert.

Thesis Table 5.13, 50,000 positions. Dashed: base CE 0.6303. Both centroid proxy and oracle need the unavailable teacher target.

Longer fine-tuning helps a weaker base

Follow-up experiment

BASE NNUE W64 / H128

0.6537

Test CE before specialization

Single evaluation head.

FOUR L1 EXPERTS

0.6499

Test CE after longer fine-tuning

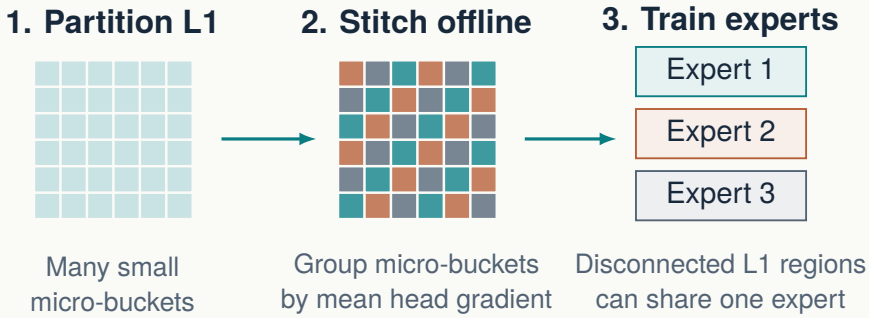
Linear dispatcher, Rprop optimizer and longer fine-tuning.

The gain depends on model and training budget.

Thesis §5.6, finding 5, and §6.1. Endpoints rounded as reported. This result concerns the weaker base, not the strong reference NNUE.

KITSUNE stitches micro-buckets offline

Proposed two-stage method



Gradients guide specialization during training.

Thesis §3.3.5. Conceptual schematic, not measured geometry. Cell colors denote the final expert assignment.

Online inference needs no gradients

KITSUNE deployment path



>99%

Micro-bucket recovery reported in the thesis.

Preliminary evidence for routing feasibility.

End-to-end benefit remains open

Final CE, memory footprint, device throughput and chess Elo need validation.

Thesis §3.3.5 and Chapter 7. Linear-router cost depends on the L1 width and micro-bucket count. Lookup has fixed cost.

The next test: end-to-end KITSUNE

Measured results and remaining validation

Established	Gradient buckets are hard to recover from L1. Strong-base MoE gains remain marginal. Longer fine-tuning improves the weaker base.
Preliminary	KITSUNE micro-bucket recovery exceeds 99%.
Next test	KITSUNE vs. strong NNUE with matched training budgets. Measure CE, memory, nodes per second and chess Elo.

Thesis Chapters 5–7. Cluster recovery, model-quality improvement and playing strength are separate validation targets.

MAIN CONCLUSION

Useful experts need a cheap, reliable assignment.

KITSUNE combines online L1 routing with offline gradient-based assignment.

Thank you. Questions?

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